INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES

Subject Code : 23A0360Se

UNIT I

Electric Vehicle Propulsion and Energy Sources

I. Introduction to Electric Vehicles

An **Electric Vehicle (EV)** is a vehicle that uses one or more **electric motors** for propulsion, with energy stored in a battery (or fuel cell / ultracapacitor) instead of burning fuel in an engine only.

Classification of EVs

- **BEV** (Battery EV) - only battery + motor, zero tail-pipe emission. Ex: Nissan Leaf, Tesla.
- **HEV** (Hybrid EV) - IC engine + electric motor, battery charged by engine / regenerative braking.
- **PHEV** (Plug-in Hybrid) - HEV with larger battery that can be charged from the grid.
- **FCEV** (Fuel-cell EV) - hydrogen fuel cell produces electricity for the motor. Ex: Toyota Mirai.

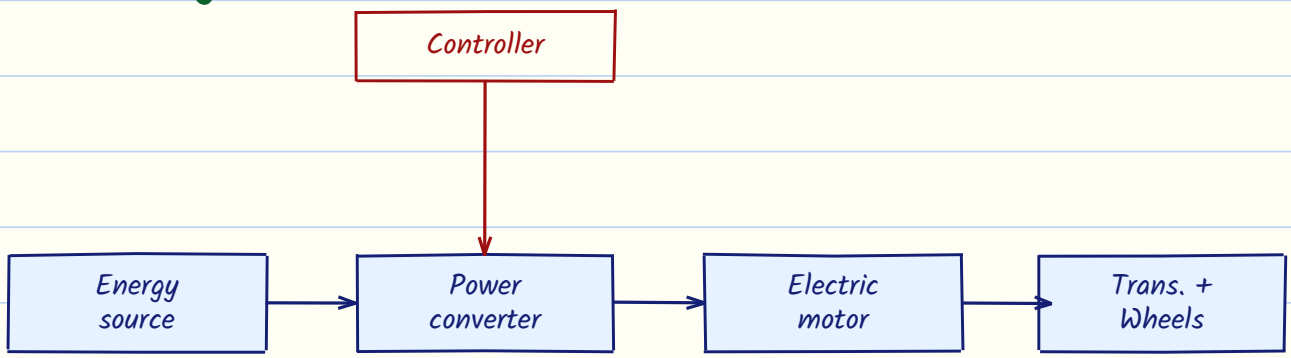
Advantages

- High efficiency (motor 85-95% vs engine 25-35%), no local pollution, low noise.
- Regenerative braking recovers kinetic energy; instant torque; low running and maintenance cost.

Limitations

- Low specific energy of batteries -> limited range; long charging time; high initial cost; charging infrastructure needed.

Block diagram of an electric drive train



Basic EV drive train (energy flows left to right; regen flows back)

The **energy source** (battery) supplies power, the **power converter** (inverter / DC-DC) controls voltage and current, the **motor** converts electrical to mechanical power, and the **transmission** (single-speed gear) drives the wheels. The **controller** takes accelerator / brake commands and drives the converter.

2. Vehicle Mechanics - Kinetics and Dynamics

Kinetics deals with the forces causing motion; **dynamics** deals with how the vehicle moves under those forces (Newton's second law). For a vehicle moving along a road, the tractive force from the wheels must overcome all the resistive forces.

Newton's second law applied to the vehicle

$$m\,(dv/dt) = \Sigma F_t - \Sigma F_r$$

where ΣF_t = total tractive force at the driven wheels and ΣF_r = sum of the resistive forces (rolling, aerodynamic, grade). Here m = vehicle mass (kg), v = speed (m/s).

Tractive force at the wheel

$$F_t = T_m\, G\, \eta_t / r_w$$

- T_m = motor torque (N-m), G = overall gear ratio, η_t = transmission efficiency, r_w = wheel radius (m).
- Wheel speed $\omega_w = v / r_w$ and motor speed $\omega_m = G\,\omega_w = G\,v / r_w$.
- Tractive force is limited by adhesion: $F_{t,max} = \mu_a\, N$, where N = normal load on driven wheels and μ_a = tyre-road adhesion coefficient (about 0.8 dry road, 0.2-0.3 wet, 0.1 ice).

3. Roadway Fundamentals

The road affects the vehicle through its **surface (rolling resistance)** and **gradient (slope)**.

(a) Rolling resistance F_{rr}

Caused by hysteresis (deformation) of the tyre and the road surface. It is almost independent of speed at low speeds.

$$F_{rr} = \mu_{rr}\, m\, g\, \cos\alpha$$

- μ_{rr} = rolling resistance coefficient: 0.008-0.015 for car tyres on concrete / asphalt, about 0.03-0.04 on gravel, 0.2-0.3 on sand.
- Depends on tyre pressure, tyre material and temperature;

low-rolling-resistance tyres are used in EVs.

(b) Aerodynamic drag F_{ad}

$$F_{ad} = \tfrac{1}{2}\,\rho\, C_D\, A_f\, (v + v_w)^2$$

- ρ = air density (1.2 kg/m³), C_D = drag coefficient (0.25–0.4 for cars), A_f = frontal area (m²), v_w = head-wind speed.
- Drag grows with the **square of speed**, so drag power grows with the cube of speed – dominant on highways.

(c) Grading (hill climbing) resistance F_{hc}

$$F_{hc} = m\, g\, \sin\alpha$$

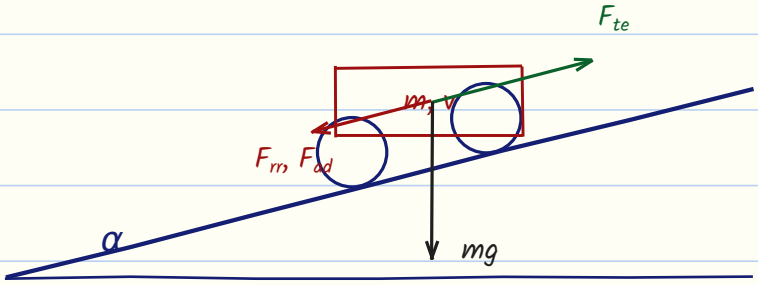
- α = road slope angle. Grade (%) = 100 tanα. For small angles sinα ≈ tanα, so 5% grade means sinα ≈ 0.05.
- Positive on uphill (resists), negative on downhill (assists motion).

(d) Acceleration force F_{la} and rotational inertia

$$F_{la} = m\, a\ ;\ F_{\omega a} = (J/r_w^2)\, a$$

The rotating parts (motor rotor, gears, wheels) also need to be accelerated. This is included by a **rotational inertia factor** $\delta = 1 + (J_{eq}/(m\, r_w^2))$ giving $F_{la} = \delta\, m\, a$, where $\delta \approx 1.05$–1.1.

Resistive forces on a slope:
rolling $F_{rr} = \mu_{rr}\, m\, g \cos\alpha$
aerodynamic $F_{ad} = \tfrac{1}{2}\, \rho\, C_D\, A_f\, v^2$
hill climb $F_{hc} = m\, g \sin\alpha$
acceleration $F_{la} = m\, a$



4. Propulsion System Design – Force-Velocity Characteristics

Total tractive effort

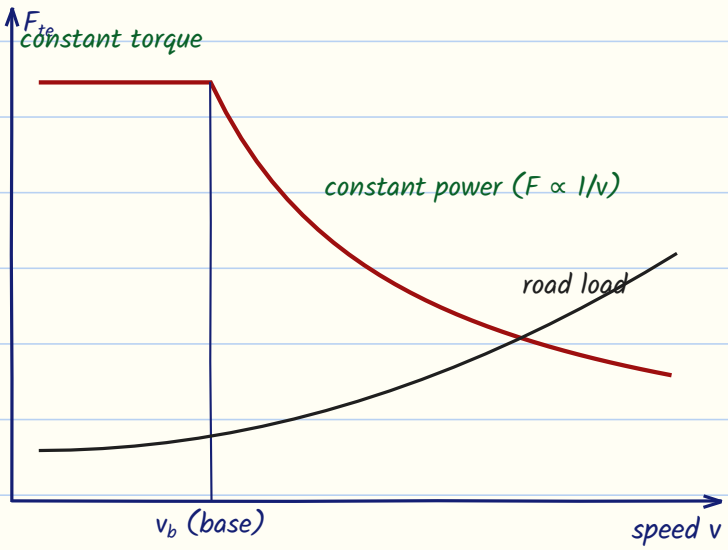
$$F_{te} = F_{rr} + F_{ad} + F_{hc} + F_{la} + F_{\omega a}$$

Propulsion design means choosing the motor, gear ratio and battery so that the EV meets the specified **top speed, acceleration (0–100 km/h) and gradeability**.

Ideal force-velocity characteristic

An ideal traction motor gives **constant torque** from zero speed up to the base speed v_b, and **constant power** beyond it. In the constant-torque region the force is constant; in the constant-power region $F = P / v$, so force falls hyperbolically with speed.

$$P_{te} = F_{te}\, v$$



Intersection of the two curves = maximum speed of the EV.

Gap between curves = force left for acceleration and gradeability.

- A multi-speed gearbox is not required for electric motors with wide constant-power range - single speed reduction ($G = 8\text{-}10$) is enough.
- The constant-power region ratio $x = \omega_{max}/\omega_{base}$ is typically 2-4 for induction/PM machines. Larger x means smaller motor torque rating for the same power.
- The gear ratio is chosen from top speed: $G = \omega_{m,max}\, r_w / v_{max}$.

5. Calculation of Tractive Power and Energy

Tractive power and energy formulas

$$P_{te}(t) = F_{te}(t) \cdot v(t)$$

$$E = \int P_{te}\, dt \text{ (traction energy at wheels)}$$

- Battery power $P_{batt} = P_{te} / (\eta_t\, \eta_m\, \eta_c)$ during motoring; during regeneration the efficiencies multiply instead ($P_{batt} = P_{te}\, \eta\, \eta\, \eta$).
- Energy consumption is expressed in Wh/km = E / distance. Range = usable battery energy / Wh per km.

Worked example 1 – cruising on flat road

Data: $m = 1500$ kg, $\mu_{rr} = 0.012$, $\rho = 1.2$, $C_D = 0.3$, $A_f = 2.2$ m², $v = 100$ km/h $= 27.78$ m/s, $\alpha = 0$.

- $F_{rr} = 0.012 \times 1500 \times 9.81 =$ **176.6 N**
- $F_{ad} = 0.5 \times 1.2 \times 0.3 \times 2.2 \times (27.78)^2 =$ **305.5 N**
- $F_{te} = 176.6 + 305.5 = 482$ N -> $P_{te} = 482 \times 27.78 =$ **13.4 kW**
- Energy for 10 km (time 0.1 h) $= 13.4 \times 0.1 =$ **1.34 kWh**, i.e. about 134 Wh/km at wheels.

Worked example 2 – climbing 5% grade at 60 km/h

- $v = 16.67$ m/s; $F_{rr} = 176.6$ N; $F_{ad} = 0.396 \times (16.67)^2 = 110$ N; $F_{hc} = 1500 \times 9.81 \times 0.05 = 735.8$ N
- $F_{te} = 1022$ N -> $P_{te} = 1022 \times 16.67 =$ **17.0 kW**

Worked example 3 – acceleration

- 0 to 100 km/h in 10 s on flat road -> $a = 27.78/10 = 2.78$ m/s²; $F_{la} = 1500 \times 2.78 = 4167$ N
- Peak power at the end of acceleration (ignoring δ) $= (4167 + 482) \times 27.78 \approx$ **129 kW**. So a motor of about 130 kW peak is needed.

> Remember: rolling resistance dominates at low speed, aerodynamic drag at high speed, and gradient/acceleration force decides the peak power rating of the motor.

6. Electric Vehicle Power Source - Battery Parameters

Terminology

- **Battery capacity (Q)** - total charge the battery can deliver, in ampere-hours (Ah), measured at a stated C-rate, temperature and cut-off voltage. Energy capacity (Wh) = $Q \times V_{nominal}$.
- **C-rate** - discharge current relative to capacity: $I = (C\text{-rate}) \times Q$. 1C for a 50 Ah cell = 50 A; 0.5C = 25 A.
- **State of Charge (SOC)** - remaining capacity as % of rated capacity.

$$SOC = (Q_{remaining} / Q_{rated}) \times 100\ \%$$
$$DOD = 100\ \% - SOC \text{ (Depth of Discharge)}$$

- **State of Discharge** - the same quantity seen from the other side: amount of charge already removed (DOD). SOC = 100% means fully charged, DOD = 100% means empty.
- **Specific energy** (Wh/kg) - energy stored per unit mass; decides the **range**. **Energy density** is per unit volume (Wh/L).
- **Specific power** (W/kg) - maximum power delivered per unit mass; decides **acceleration** and gradeability.
- **Terminal voltage, internal resistance, cycle life, self-discharge, efficiency** (Ah and Wh efficiency).

Peukert's relation (effect of discharge rate on capacity)

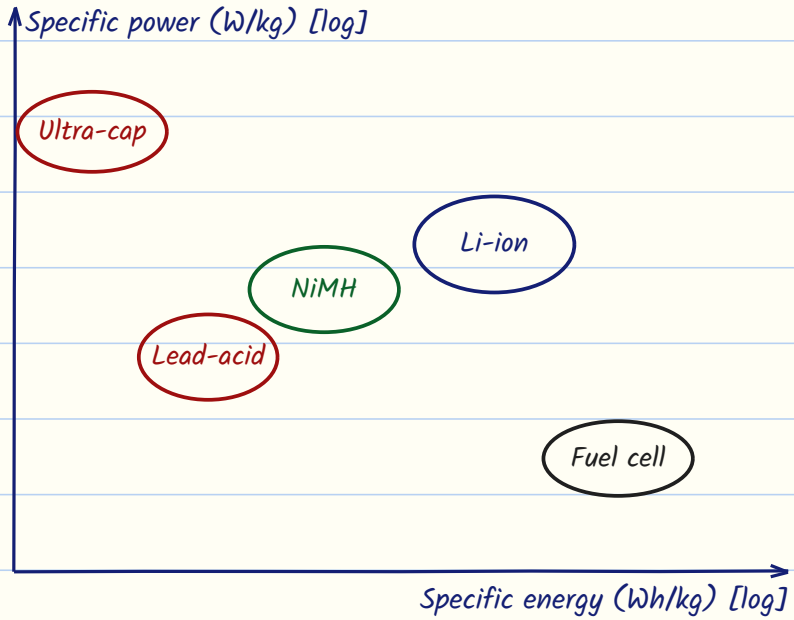
$$t = H\,(\,C / (I\,H)\,)^{k}$$

H = rated discharge time (h), C = rated capacity at H hours, I = actual current, k = Peukert constant (1.05-1.3 for lead-acid, about

1.0-1.1 for Li-ion). Higher current -> less available capacity.

Ragone plot

A **Ragone plot** is a log-log graph of **specific power (y-axis) against specific energy (x-axis)** for energy storage devices. It lets us compare sources: a device cannot give both very high energy and very high power. EVs need both, so batteries are often combined with ultracapacitors.



Ragone plot:
shows power vs energy trade-off of each source. Top-left: high power, low energy (ultracap). Bottom-right: high energy, low power (fuel cell).

7. Battery Modeling

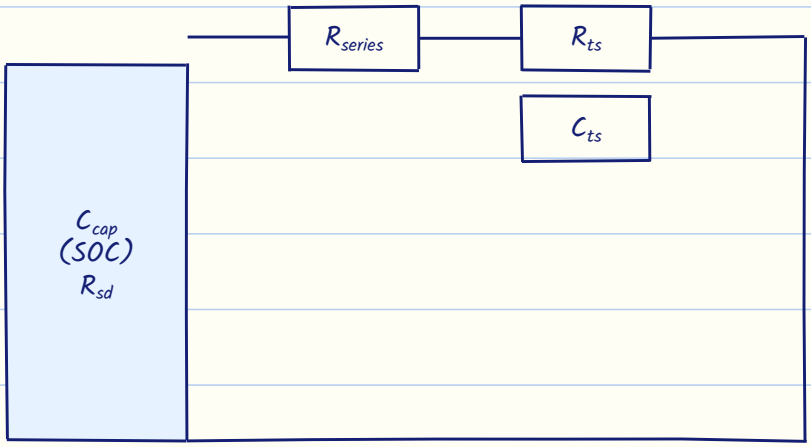
A battery model predicts terminal voltage, SOC and run time for a given load current. Used in simulation, BMS and EV range estimation.

(a) Run-time battery model

An electrical-circuit (equivalent circuit) model. It predicts both the **transient response** and the **run time** (how long the battery lasts) for a given load. Components: a **capacitor C_{cap}** whose

voltage represents SOC, a self-discharge resistor R_{sd}, a voltage source $V_{oc}(SOC)$, series resistance R_{series} and RC networks for transients. Parameters are fitted from discharge tests and depend on SOC, temperature and cycle number.

Run-time battery model (Thevenin / Chen-Rincon-Mora type)



$V_{oc}(SOC)$: open circuit
R_{series} : ohmic drop
R||C : transient (short term)
C_{cap} : stored charge = SOC
R_{sd} : self-discharge

- Run time = time taken for the SOC voltage on C_{cap} to fall to the cut-off value under the load current.
- Simple, fast, easy to put in vehicle simulations (MATLAB / Simulink, ADVISOR).

(b) First-principle (electrochemical) model

Based on the physics and chemistry inside the cell: electrode reactions, ion diffusion in electrolyte and solid particles and charge conservation (porous electrode theory, Doyle-Fuller-Newman model).

- Open-circuit voltage from the **Nernst equation**: $E = E_0 - (R\,T / n\,F) \ln Q_r$.
- Reaction kinetics - **Butler-Volmer equation**: $i = i_0[\exp(\alpha_a F \eta$

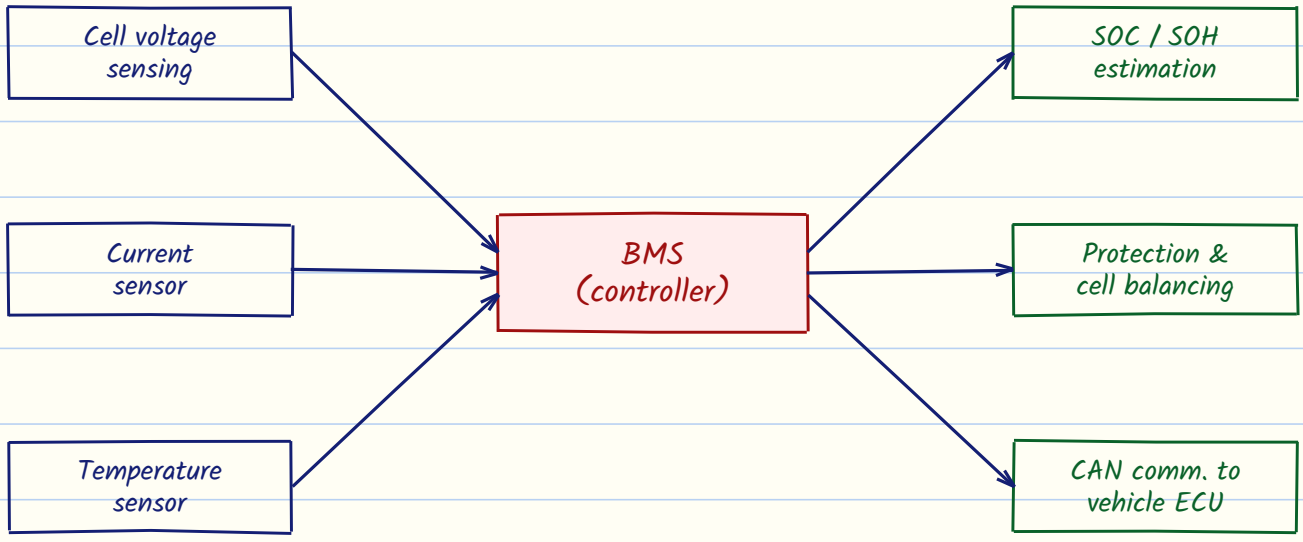
$/ R T) - \exp(-\alpha_c F \eta / R T)]$, where η = overpotential.

- Diffusion of lithium ions in the solid and electrolyte - **Fick's law**: $\partial c/\partial t = D \nabla^2 c$.
- Very accurate and gives internal states (concentration, temperature) but needs many parameters and heavy computation - used for cell design, not in real-time BMS.

Feature	Run-time (circuit) model	First-principle model
Basis	Electrical analogy, curve fitting	Electrochemistry, physics
Accuracy	Moderate	High
Computation	Light, real-time	Heavy
Use	BMS, vehicle simulation	Cell design / research

8. Battery Management System (BMS)

The **BMS** is the electronic system that monitors and protects the battery pack and keeps it in its safe operating area.



Functions of BMS

- Measures cell voltages, pack current, temperatures; protects against over-charge, over-discharge, over-current and over/under temperature.
- Estimates **SOC** (fuel gauge) and **SOH** (state of health), calculates remaining range and available power.
- Cell balancing, thermal management (fan / liquid cooling control), contactor control and fault diagnosis; communicates on CAN bus.

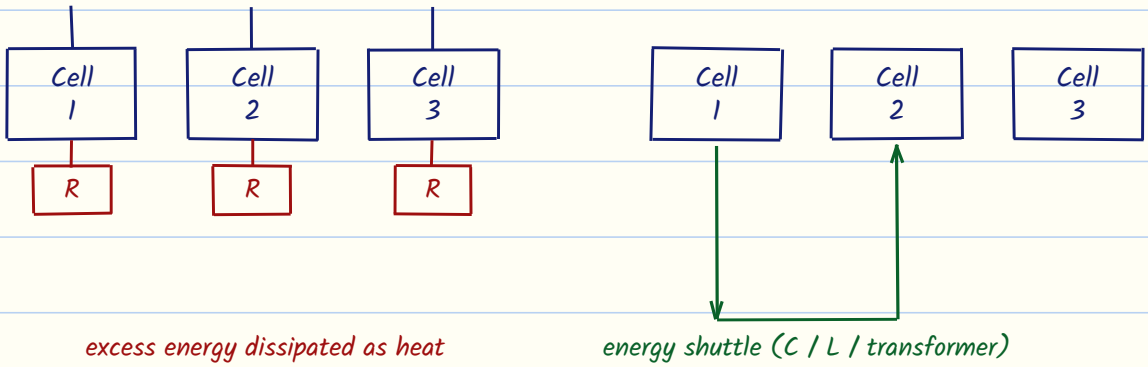
SOC measurement methods

- **Coulomb counting (ampere-hour integration)**: $SOC(t) = SOC(0) + (1/Q_n) \int \eta\, I\, dt$ (I positive on charge). Simple but drifts due to current-sensor error and unknown initial SOC.
- **Open-circuit voltage method**: SOC read from the V_{oc}-SOC curve after the battery rests. Accurate but needs rest time; flat curve for LiFePO4 / NiMH makes it hard.
- **Impedance / internal-resistance method** and (for lead-acid) specific gravity of electrolyte.
- **Kalman filter / observer-based method**: combines model and measured V, I to give an optimal SOC estimate that corrects drift - used in modern EVs.
- Practical BMS uses coulomb counting + OCV correction + Kalman filter together.

Battery cell balancing

Cells in a series string differ slightly in capacity, internal resistance and self-discharge, so some cells reach full / empty earlier. The weakest cell then limits the pack. **Cell balancing** equalises the SOC of all cells.

(a) Passive balancing (b) Active balancing



- **Passive balancing** - a bleed resistor across the higher-voltage cell burns the excess charge as heat. Cheap, simple but wasteful; used mostly during charging.
- **Active balancing** - transfers charge from strong cells to weak cells using capacitors (switched capacitor), inductors (buck-boost) or transformers. Efficient, faster, costlier and more complex.

9. Traction Batteries

(a) Nickel-Metal Hydride (NiMH) battery

- Positive electrode: nickel oxyhydroxide NiOOH; negative: hydrogen-absorbing metal alloy (MH); electrolyte: KOH. Nominal cell voltage **1.2 V**.

- Reaction: $NiOOH + MH <-> Ni(OH)_2 + M$ (charge <-> discharge). No cadmium, so environment-friendly.
- Specific energy 60-120 Wh/kg, specific power 200-1000 W/kg, life 500-1500 cycles, high self-discharge, good safety, wide temperature range.
- Used in Toyota Prius (first and second generation) and many hybrid cars.

(b) Lithium-Ion (Li-ion) battery

- Cathode: lithium metal oxide (LCO, NMC, NCA, LFP); anode: graphite; electrolyte: lithium salt ($LiPF_6$) in organic solvent; separator between. Nominal voltage 3.2-3.7 V.
- During charge Li^+ ions move from cathode to anode (intercalate in graphite); on discharge they return and electrons flow through the external circuit.
- Specific energy 100-265 Wh/kg, specific power 300-1500 W/kg, life 1000-3000 cycles, low self-discharge (2-5% per month), no memory effect, efficiency above 95%.
- Needs BMS: overcharge / over-temperature can cause thermal runaway.

(c) Lithium-Polymer (Li-poly) battery

- A Li-ion variant that uses a **gel / solid polymer electrolyte** instead of liquid; cells are thin flat pouch type - flexible shape and light weight.
- Specific energy 130-200 Wh/kg, lower risk of leakage, better

packaging, but lower conductivity at low temperature and higher cost; life 500-1000 cycles.

Comparison

Parameter	Lead-acid	NiMH	Li-ion	Li-poly
Cell voltage (V)	2.0	1.2	3.6	3.7
Specific energy (Wh/kg)	30-40	60-120	100-265	130-200
Specific power (W/kg)	~180	200-1000	300-1500	~250-2000
Cycle life	200-500	500-1500	1000-3000	500-1000
Self discharge	Low	High (~30%/mo)	Very low	Very low
Cost	Lowest	Medium	High	Highest

Important questions: (1) Derive the tractive effort equation. (2) Explain Ragone plot. (3) SOC estimation methods. (4) Cell balancing. (5) Compare NiMH, Li-ion and Li-poly batteries.